



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM - 638 401

Module Test - I - Apr - 2026

II Semester

22GE004 & BASICS OF ELECTRONICS ENGINEERING

1	(i)	A factory production line uses an electrically powered conveyor system to move heavy materials between processing stations. Over time the conveyor motor begins consuming higher electrical power while producing the same mechanical output speed. Maintenance engineers suspect inefficiencies in the energy transfer process within the system. Electrical measurements show increased current while mechanical output remains constant. The team must diagnose the underlying cause affecting energy flow. Analyze the situation and infer the most plausible energy transfer issues causing the observed increase in power consumption. (3 Marks - [U/C, 2])
2	(ii)	An automated irrigation system uses solar panels to power a small electric pump that lifts water from a reservoir to agricultural fields. Solar radiation generates electrical energy in the panels which is then transmitted through wires to operate the pump motor. The motor converts this electrical energy into mechanical motion that drives the pumping mechanism. The system therefore involves multiple stages of energy transformation before water is delivered to the crops. Infer the logical sequence of energy transformations occurring within the irrigation system architecture. (2 Marks - [U/C, 1])
3	(i)	An industrial automation company installs sensors across a manufacturing line to monitor pressure, temperature, and machine vibration levels. Engineers realize that raw sensor outputs differ widely in voltage level, impedance, and noise characteristics. Without proper signal conditioning, the measurement system could misinterpret these signals and trigger incorrect control actions in the production line. Therefore, the engineering team must evaluate the overall importance of signal conditioning within the measurement architecture before deploying the system in a real industrial environment. Assess the professional significance of signal conditioning in industrial measurement systems and recommend the critical functions it performs to ensure reliable monitoring and control. (3 Marks - [U/C, 2])
4	(ii)	A communication system processes signals from multiple transmitters using a centralized receiver module. Engineers must determine whether adding dedicated signal conditioning circuits for each transmitter improves performance or increases system complexity unnecessarily. The design decision must balance signal quality, circuit efficiency, and scalability when expanding the communication network. Infer how signal conditioning improves communication reliability while reconciling the efficiency trade-offs associated with adding conditioning circuits in large signal processing systems. (2 Marks - [R/F, 1])

5	(i)	A smart grid project integrates multiple renewable energy sources such as solar panels and micro wind turbines into a local power distribution network. Energy from these sources must travel through transmission lines before reaching residential consumers. Engineers must determine how energy carriers and transmission pathways influence system efficiency. Increasing transmission distance improves grid coverage but introduces energy losses. The planning team must analyze this trade-off before expanding the system. Evaluate how transmission pathways influence energy efficiency in the smart grid and determine the critical engineering considerations. (3 Marks - [U/C, 2])
6	(ii)	A technology company is designing next-generation wearable electronic devices that rely on compact batteries and miniature circuits to operate continuously throughout the day. Engineers must ensure that the limited stored energy inside the device is transmitted efficiently to sensors, processors, and wireless transmitters. Any inefficiency in the internal energy pathways will reduce battery life and device reliability. The development team therefore evaluates the role of energy transfer principles in system design. Assess why efficient energy transfer is a critical engineering requirement in such compact technological systems. (2 Marks - [U/C, 1])
7	(i)	A wireless communication laboratory compares three electromagnetic signal transmission technologies: radio communication used in broadcasting, microwave transmission used in radar systems and optical signals used in fiber-optic internet infrastructure. Researchers analyze differences in wavelength, frequency and propagation mechanisms to determine how each technology supports distinct communication functions. The evaluation aims to understand how electromagnetic signal characteristics influence transmission applications. Integrate electromagnetic wave characteristics with communication technology requirements and determine how wavelength and frequency relationships guide the selection of radio, microwave or optical signal transmission methods. (3 Marks - [U/C, 2])
8	(ii)	A technology firm compares microwave satellite communication with optical fiber transmission for delivering high-speed internet between two cities. Microwave links require antennas and free-space propagation while optical systems require fiber infrastructure but provide higher bandwidth. Engineers analyze scalability and efficiency of electromagnetic signal transport under these conditions. Analyze the scalability trade-offs between microwave and optical electromagnetic transmission methods and infer which system offers higher long-term data capacity. (2 Marks - [U/C, 1])
9	(i)	A rotating steel shaft in a processing plant transmits mechanical power from a turbine to a milling machine. The shaft rotates at 600 rpm while delivering a torque of 200 N·m. Engineers must evaluate the mechanical power transmitted through the shaft to ensure the machine operates within design limits. The evaluation also determines whether structural reinforcement is required for the shaft assembly. Determine the mechanical power transmitted through the shaft and interpret the result to assess the effectiveness of the energy transfer mechanism. (5 Marks - [Ap/P, 3])

10	(i)	A biomedical monitoring device measures human heartbeat activity and converts the mechanical vibration of the chest into an electrical signal for digital analysis. Engineers observe that the signal amplitude varies periodically with each heartbeat, and the system records the signal both in the time domain and frequency domain to identify abnormal rhythms. The analysis software detects frequency components corresponding to periodic cardiac activity. Engineers want to understand how signal representation enables both energy transmission and information interpretation within the monitoring system. Formulate how the heartbeat signal can be interpreted through time-domain and frequency-domain representations to extract physiological information. Analyze how these signal representations assist engineers in recognizing periodic patterns and interpreting energy variations associated with cardiac activity. (5 Marks - [Ap/P, 3])
11	(i)	A conveyor system in a mining facility uses a gear transmission to transfer power between shafts. After several months of operation, engineers observe overheating near the gearbox and a gradual drop in mechanical efficiency. Inspection reveals increased vibration and metallic noise during operation. The maintenance team suspects energy losses caused by mechanical resistance within the transmission system. Analyze the operational symptoms and determine the most probable mechanical causes of energy loss in the gear transmission system. Infer corrective measures based on mechanical transmission principles. (5 Marks - [Ap/P, 3])
12	(i)	An automated irrigation system in a large agricultural field uses soil moisture sensors connected to a central monitoring unit. The sensors produce electrical signals whose amplitude changes according to soil moisture levels. The monitoring unit continuously interprets these signals to decide when irrigation pumps must be activated. Engineers observe that although electrical power reaches the monitoring unit correctly, incorrect irrigation decisions sometimes occur due to fluctuations in the transmitted signal. The design team is analyzing how signals simultaneously carry energy and information within this sensing network. Analyze how the electrical signal in the irrigation monitoring network simultaneously functions as a carrier of energy and encoded information. Determine how variations in signal amplitude enable the monitoring system to interpret soil conditions and justify the role of signal integrity in ensuring accurate irrigation decisions. (5 Marks - [Ap/P, 3])
13	(i)	A manufacturing plant distributes electrical energy through long transmission cables to robotic assembly stations. Recently, several stations intermittently fail to respond to digital control signals transmitted as small voltage variations along the same electrical lines. Measurements reveal higher than expected resistance in certain cable segments due to aging insulation and conductor corrosion. Engineers must diagnose why both electrical power delivery and signal communication are degrading within the network. A systematic electrical transmission analysis is required before replacing equipment. Investigate the electrical transmission network and infer the mechanism by which increased conductor resistance simultaneously disrupts energy delivery and signal transport. Justify a technically reasoned corrective approach that restores efficient electrical transmission within the plant. (5 Marks - [Ap/P, 3])

14	(i)	A coastal weather station sends atmospheric data to a research center using an electromagnetic radio communication system. Recently the system has experienced delayed signal reception and reduced signal strength during data transmission, particularly when large datasets are transmitted at high frequency intervals. Engineers suspect that bandwidth limitations and environmental interference are affecting the signal propagation and transmission reliability across the long communication path. Diagnose the probable technical factors responsible for the observed signal degradation and transmission delays in the communication system. Interpret how bandwidth constraints, attenuation and propagation distance collectively influence signal transmission performance in this scenario. (5 Marks - [Ap/P, 3])
15	(i)	An automated greenhouse complex distributes electrical energy from a central solar inverter to multiple climate-control subsystems using copper conductors arranged in radial circuits. Engineers observe that distant irrigation controllers receive lower voltage compared with units closer to the inverter because of resistance along the conductors. Sensors also transmit control signals as voltage variations across the same network. The design team must decide whether to redesign the conductor layout to maintain reliable power and signal transport. The objective is to ensure stable electrical energy transmission and accurate signal propagation across the entire facility. Analyze the electrical transmission situation and determine how the interaction of current, voltage, and resistance influences both power delivery and signal transport in the greenhouse network. Formulate a technically justified redesign strategy that improves transmission reliability while preserving signal integrity across distant subsystems. (5 Marks - [Ap/P, 3])
16	(i)	A robotic manufacturing line uses mechanical motion sensors attached to rotating shafts to detect equipment vibrations. The mechanical signals generated by these sensors must be transmitted to a digital control system that analyzes machine health in real time. Engineers install transducers that convert mechanical vibrations into electrical signals before sending them through electronic circuits to the monitoring computer. Determine the engineering rationale behind introducing transducers in the robotic monitoring system and recommend how mechanical-to-electrical signal conversion improves transmission efficiency and data processing capability in this industrial setup. (5 Marks - [Ap/P, 3])
17	(i)	Engineers testing early audio amplifiers noticed that diode-based rectifiers produced signal distortion during high-frequency oscillations. They needed to adjust the circuit to minimize waveform alteration while maintaining voltage rectification efficiency. Question: Infer how the engineers should modify the vacuum tube configuration to reduce distortion. (3 Marks - [U/C, 2])
18	(ii)	A telegraph system experienced signal fading due to insufficient electron flow in its vacuum tube amplifiers. Engineers needed a simple adjustment to enhance conduction without redesigning circuits. Question: Recommend a quick operational adjustment to improve signal strength. (2 Marks - [U/C, 1])

19	(i)	A designer used a PN junction diode in a signal demodulation circuit and needed to ensure that forward conduction occurred efficiently while reverse bias prevented backflow. Question: Prioritize design considerations to optimize diode operation in signal processing. (3 Marks - [U/C, 2])
20	(ii)	A rectifier circuit experienced inefficient energy conversion due to improper diode selection. Engineers had to adjust diode choice to improve AC to DC conversion efficiency. Formulate a method to enhance rectification efficiency using diode characteristics. (2 Marks - [U/C, 1])
21	(i)	During a power regulation study, technicians observed that electron emission from vacuum tube cathodes varied with ambient temperature, affecting amplifier stability. They needed a method to maintain consistent current flow despite environmental changes. Question: Prioritize design adjustments to stabilize vacuum tube operation under varying temperatures. (3 Marks - [U/C, 2])
22	(ii)	Early experimental amplifiers suffered from inconsistent gain due to uncontrolled electron flow across the triode vacuum tubes. Technicians had to improve circuit stability with minimal component changes. Question: Formulate a method to stabilize triode operation efficiently. (2 Marks - [U/C, 1])
23	(i)	During a protective circuit test, engineers noticed that applying a reverse voltage across a PN junction diode resulted in very small leakage current. They needed to quantify how built-in potential and minority carriers limited current under reverse bias. Infer the reasons for minimal current flow under reverse voltage. (3 Marks - [U/C, 2])
24	(ii)	Engineers noted that in a protection circuit, a PN junction diode prevented voltage spikes from damaging sensitive components. Recommend a practical method to implement the diode for unidirectional protection. (2 Marks - [U/C, 1])
25	(i)	An experimental high-frequency radio transmitter in 1950s used vacuum tubes, but switching speeds were insufficient for rapid signal modulation. Engineers considered alternative devices to achieve high-speed electronic control for broadcasting. Question: Recommend a strategy to overcome the speed limitations of vacuum tubes for high-frequency applications. (5 Marks - [Ap/P, 3])
26	(i)	Engineers were tasked with creating amplitude-multiplied signals in a communication receiver using diode networks. The system required precise waveform shaping, so the network needed correct diode placement, biasing, and configuration to avoid distortion. They also had to consider current handling, voltage drops, and linearity to ensure the multiplied signal matched theoretical expectations for amplitude while maintaining fidelity in downstream processing circuits. Recommend a PN junction diode network configuration for signal multiplication and waveform control. (5 Marks - [Ap/P, 3])

27	(i)	A communication satellite prototype relied on vacuum tube amplifiers but experienced high power consumption and mechanical fragility during vibration tests. The engineers needed to evaluate alternatives that could provide the same amplification functionality with reduced size, weight, and power requirements. Question: Analyze the factors guiding the transition from vacuum tubes to semiconductor devices for satellite applications. (5 Marks - [Ap/P, 3])
28	(i)	: A precision measurement system received AC signals that needed shifting in DC voltage to match the input range of subsequent circuitry. The engineers could not tolerate waveform distortion as it would affect measurement accuracy. They considered designing a diode-based clamping circuit with capacitors and appropriate biasing to shift the voltage level while keeping the original AC waveform intact and ensuring the system's signals stayed within safe operating levels. Analyze how PN junction diodes can be applied for signal clamping in measurement systems. (5 Marks - [Ap/P, 3])
29	(i)	A research team investigated the effect of temperature on intrinsic and extrinsic semiconductors for thermistor development. They observed that intrinsic materials showed exponential conductivity changes, whereas doped semiconductors had moderated response. They needed to formulate a material selection and doping plan to achieve predictable temperature-dependent behavior. Recommend a semiconductor type and doping level suitable for stable thermistor operation. (5 Marks - [Ap/P, 3])
30	(i)	Engineers were designing a high-frequency rectifier circuit using PN junction diodes and observed that reverse leakage current increased significantly at elevated temperatures. The diode's switching speed and junction capacitance also limited the maximum frequency at which the circuit could operate efficiently. The team needed to analyze how these performance constraints affected the rectifier output, including waveform distortion and efficiency loss, and determine possible design adjustments or alternative diode choices for optimal operation. Determine the key performance limitations of PN junction diodes in high-frequency rectifiers and recommend strategies to mitigate them. (5 Marks - [Ap/P, 3])
31	(i)	An electronics company needed to manufacture a precision diode that required controlled conduction. Engineers had to differentiate between conductors, insulators, and semiconductors based on band gap and available charge carriers. They also had to select suitable doping strategies to achieve targeted n-type or p-type behavior for circuit integration. Analyze the classification criteria and propose the doping strategy to produce the desired diode characteristics. (5 Marks - [Ap/P, 3])
32	(i)	In a power supply design, engineers observed that PN junction diodes approached their breakdown voltage under peak load conditions. They needed to consider junction voltage limitations, temperature effects, and power dissipation to ensure reliability. They also evaluated whether advanced semiconductor devices could provide higher voltage tolerance while preserving rectification efficiency. Recommend design considerations to prevent diode failure due to voltage and thermal constraints. (5 Marks - [Ap/P, 3])

33	(i)	A research engineer designs a measurement system that collects electrical signals from multiple sensors connected to a central processing unit. The system must ensure that signals are correctly amplified, cleaned from noise, and routed to the correct processing channels. Improper ordering of these signal processing stages could distort measurements or overload circuit components. Therefore, the engineer must determine the logical flow of signal conditioning operations within the instrumentation architecture before signals reach the processing system. Interpret the logical sequence of signal conditioning operations required in an instrumentation system and reconcile how each stage prepares the electrical signal for reliable processing. (3 Marks - [U/C, 2])
34	(ii)	A laboratory technician notices that a temperature measurement system produces fluctuating readings even though the environmental temperature remains constant. The sensor itself appears functional, but the signal reaching the processing unit shows irregular voltage variations. Engineers suspect that interference and poor signal conditioning may be affecting the measurement accuracy. Diagnose the likely cause of the unstable readings by interpreting how inadequate signal conditioning could distort the electrical signal obtained from the sensor. (2 Marks - [R/F, 1])
35	(i)	A radio communication device requires an amplifier capable of increasing signal strength without altering the original signal shape. Engineers analyze factors that affect amplifier performance including distortion and frequency response. Assess the influence of frequency response and distortion on the performance of transistor amplifier circuits. (3 Marks - [U/C, 2])
36	(ii)	A common-emitter amplifier circuit produces a distorted output signal where only the positive half cycles of the input signal are visible, while the negative half cycles are clipped. On investigation, it is found that the transistor is operating in the saturation region due to excessive base biasing. Diagnose the reason for the distorted output and interpret the condition required for proper amplification. (2 Marks - [R/F, 1])
37	(i)	An industrial automation company installs sensors across a manufacturing line to monitor pressure, temperature, and machine vibration levels. Engineers realize that raw sensor outputs differ widely in voltage level, impedance, and noise characteristics. Without proper signal conditioning, the measurement system could misinterpret these signals and trigger incorrect control actions in the production line. Therefore, the engineering team must evaluate the overall importance of signal conditioning within the measurement architecture before deploying the system in a real industrial environment. Assess the professional significance of signal conditioning in industrial measurement systems and recommend the critical functions it performs to ensure reliable monitoring and control. (3 Marks - [U/C, 2])

38	(ii)	A communication system processes signals from multiple transmitters using a centralized receiver module. Engineers must determine whether adding dedicated signal conditioning circuits for each transmitter improves performance or increases system complexity unnecessarily. The design decision must balance signal quality, circuit efficiency, and scalability when expanding the communication network. Infer how signal conditioning improves communication reliability while reconciling the efficiency trade-offs associated with adding conditioning circuits in large signal processing systems. (2 Marks - [R/F, 1])
39	(i)	During the design of a transistor amplifier circuit, engineers must ensure that the transistor operates in the correct region to achieve proper amplification. They analyze the biasing network consisting of resistors connected to the base and collector terminals. Interpret how biasing networks maintain stable amplifier operation in transistor circuits. (3 Marks - [U/C, 2])
40	(ii)	Modern communication systems require amplifiers capable of processing signals across different frequencies while maintaining stable performance. Infer how transistor amplifiers improve efficiency in communication and signal processing systems. (2 Marks - [R/F, 1])
41	(i)	A communication device uses a transistor to process incoming electrical signals transmitted through a wireless receiver. The circuit must operate reliably while switching between different operational states depending on the input signal strength. Engineers observe that when the input signal is absent, the transistor stops conducting current. When a moderate signal is present, the transistor amplifies the signal. When the input becomes very strong, the transistor allows maximum current flow and behaves like a closed switch. To design the system correctly, the engineers must understand the different operational modes of the transistor and how biasing conditions influence these modes. Integrate the concepts of transistor biasing and operational regions to interpret how a Bipolar Junction Transistor transitions between cutoff, active, and saturation modes in signal processing circuits. (5 Marks - [An/C, 3])
42	(i)	An electronics company is developing a compact communication device that must operate efficiently while consuming minimal power. Initially, engineers consider using Bipolar Junction Transistors for amplification and switching functions within the circuit. However, during the design review, they observe several limitations associated with BJTs such as high power consumption and sensitivity to temperature variations. As the circuit operates continuously for long durations, the engineers are concerned about thermal instability and the possibility of device failure under extreme operating conditions. They also analyze how the transistor's operating characteristics might affect the reliability and efficiency of the overall system. To ensure optimal performance, the design team must carefully evaluate the practical limitations of BJTs before deciding whether to continue using them in the circuit. Analyze the major limitations of Bipolar Junction Transistors in electronic circuits and justify why these factors must be considered during device selection. (5 Marks - [An/C, 3])

43	(i)	A laboratory experiment studies current flow in a Bipolar Junction Transistor used in a signal amplification circuit. Students measure the emitter current, base current, and collector current under normal operating conditions. During testing, they observe that the emitter current entering the transistor is divided between the base terminal and the collector terminal. The experiment aims to verify the current relationship that governs transistor operation. By analyzing measured values, students attempt to confirm whether the fundamental current relationship predicted by transistor theory holds true in practical operation. Formulate the mathematical relationship between emitter current, base current, and collector current and interpret how the conservation of charge explains the observed current distribution in the transistor. (5 Marks - [An/C, 3])
44	(i)	A research team is designing a high-frequency communication system that requires components capable of operating at very high signal frequencies. During testing, engineers notice that BJTs do not perform efficiently at extremely high frequencies due to internal capacitances between transistor junctions. These capacitances affect signal transmission and limit the transistor's ability to respond quickly to rapid signal changes. To improve system performance, engineers must analyze how these internal characteristics influence the behavior of the transistor. Integrate the concepts of junction capacitance and high-frequency operation to interpret why BJTs face limitations in high-frequency electronic circuits. (5 Marks - [An/C, 3])
45	(i)	An electronic control system uses a BJT to drive a relay that activates a high-power device such as a motor or pump. The transistor must quickly switch between ON and OFF states depending on the input signal from the control unit. Engineers analyze the switching characteristics of the transistor including the time required to transition between states and the delay between input and output response. These factors determine how effectively the transistor can control external devices in real-time systems. Integrate the concepts of switching characteristics such as switching time and propagation delay to interpret their impact on transistor switching performance in electronic control systems. (5 Marks - [An/C, 3])
46	(i)	In a semiconductor laboratory experiment, students compare the operational characteristics of a Bipolar Junction Transistor and a Field Effect Transistor used in similar amplifier circuits. They observe that the BJT requires continuous base current to control collector current, while the FET operates using gate voltage without significant gate current. By analyzing these behaviors, students attempt to understand the fundamental differences between current-controlled and voltage-controlled transistor devices. Formulate the operational differences between BJT and FET devices and interpret how voltage control in FETs improves circuit efficiency. (5 Marks - [An/C, 3])

47	(i)	An industrial automation system requires a reliable electronic switching mechanism to control the operation of motors and indicator lights in a production line. Engineers are evaluating different switching devices and decide to use a Bipolar Junction Transistor instead of mechanical switches because of its fast response and compact size. The control circuit receives digital signals from a microcontroller which determine whether the transistor should allow current to flow or block it completely. During the design process, the engineers must understand how a BJT behaves when used as a switching device and identify the operating regions required for proper switching. They also need to determine how the base current controls the switching action of the transistor so that the system responds accurately to digital signals. The design must ensure reliable ON and OFF switching without damaging the transistor or causing circuit malfunction. Analyze how a Bipolar Junction Transistor operates as an electronic switch in digital control circuits and justify the operating regions required for proper switching action. (5 Marks - [An/C, 3])
48	(i)	A semiconductor design team is developing a compact electronic device that requires efficient amplification and switching circuits. Initially, the engineers used Bipolar Junction Transistors in the design, but they observed certain limitations such as higher power consumption and lower input impedance. To improve circuit efficiency, the team considers using Field Effect Transistors instead. They study the basic structure of FET devices and understand that the current flow through the device is controlled by the voltage applied at the gate terminal rather than by current as in BJTs. The engineers also analyze how the channel between the source and drain terminals allows current flow when an appropriate gate voltage is applied. Understanding the operating principle of FET devices will help the design team select the most suitable transistor technology for the circuit. Analyze the basic operating principle of a Field Effect Transistor and justify how gate voltage controls current flow through the device. (5 Marks - [An/C, 3])